HATCHSIDE · MARKET AND VIABILITY

HatchSide market and viability

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DOCUMENT CONTROL

Document ID	HTS-MKT-001	Version	0.1
Status	Draft	Date	2026-11-05
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/hatchside		
Source	docs/10-market.md @ 4004f47		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-05	Amish Chadha	First draft at the kit 1.9.1 upgrade (stage P3), for option A

HatchSide market and viability

Municipal sanitation departments, biogas programmes and informal sanitation crews need a low-cost way to stay out of confined spaces, and the commercial tripod-and-winch sets that exist cost about USD 5,200 to 6,100 and carry no work tools; HatchSide's parts estimate is USD 2,838 and an open design lets any workshop build it. Prices and benchmarks are cited with the date they were checked (2026-11-05). Method: STANDARDS section 23.

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
Municipal sewer and drain crews	Urban local bodies in India; later other countries with mechanised sanitation programmes (NAMASTE, HTS-REG-001 G10)	Bandicoot cleaning robot: first unit about INR 10 lakh (about USD 1,500 at the time, 2018 article, Quartz); beta units INR 10 to 12 lakh to make (Forbes India); both are older prices	2,838 (BOM, option A)	A manual kit that does the grab, rake and sample jobs for a fraction of a robot, repaired in the municipal workshop
Household biogas technicians and programmes	Rural India, Nepal and similar; DomeReach with the shared frame	No close product found; digester maintenance guidance says work from above ground where possible (energypedia)	2,838 for the kit; DomeReach alone about 225 plus the share of the frame (BOM line 21)	Reseal a dome from the opening without entering
Industrial and oil and gas tank contractors	Tank roof hatches, frac tanks, in regulated markets	Confined-space tripod with winch: USD 5,853.29 to 6,120.19 across variants (PK Safety, 3M DBI-SALA 83010); a similar 7 ft set with 60 ft cable listed at USD 5,229.00 and a 7 ft tripod alone at USD 1,660.00 on retailer pages seen in search results (links not recorded; to re-check)	2,838	Same price class for a kit that carries work tools; but it is not certified for people and cannot replace a rescue tripod
Winch only (component benchmark)	Buyers and fabricators	Thern 1,000 lb worm-gear hand winch with automatic brake: USD 715.99 (GME Supply)	BOM winch line: 320 (a two-speed winch, supplier not chosen)	Shows the winch price range; the BOM figure is an estimate to confirm on purchase

Viability: on paper the kit is cheaper than the commercial sets because it is a material hoist and not a personnel rescue system, so it skips the certification that adds most of their price. That also limits who may buy it for which use (HTS-REG-001 G1 to G6). Whether crews will choose it over the habit of entering is a co-design question that paper cannot answer.

Open route

Who would build it: municipal workshops, local fabricators with a welder and drill, sanitation NGOs, university engineering departments and makerspaces. The drawings, BOM and build plan are published under CERN-OHL-S-2.0 (hardware) and MIT (software), so anyone can build and sell the design, and improvements must be shared back. A workshop that builds it for sale takes on the certification and liability questions in HTS-REG-001; the repository does not.

Sources

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- Energypedia, "Maintenance of a Biogas Plant," https://energypedia.info/wiki/Maintenance_of_a_Biogas_Plant, as cited in HTS-PRB-001.
- Retailer listings for a 3M DBI-SALA 7 ft tripod with 60 ft cable (USD 5,229.00) and a 3M 7 ft tripod alone (USD 1,660.00), seen in web search results on 2026-11-05; the pages were not recorded, so these two figures are to re-check before use.